FUNDAMENTALS OF CYBER SECURITY
MODULE 2 — HANDWRITTEN QUESTIONS & ANSWERS

Exam-ready notes based on all 9 questions listed under Module 2 in the uploaded FCSDF Question Bank.

Q1. Explain DES algorithm with block diagram.

Answer:

DES (Data Encryption Standard) is a symmetric-key block cipher. It uses the same secret key for encryption and decryption.

Main features:

- Block size: 64 bits.
- Effective key size: 56 bits (8 additional bits are used for parity).
- Structure: Feistel network.
- Number of rounds: 16.

Working of DES:

1. Initial Permutation (IP): The 64-bit plaintext is rearranged according to a fixed permutation.
2. Splitting: The result is divided into two 32-bit halves, L0 and R0.
3. Sixteen rounds: Each round uses a different 48-bit round key.

- Expansion: The 32-bit right half is expanded to 48 bits.
- XOR: The expanded value is XORed with the round key.
- S-box substitution: The 48-bit result is reduced to 32 bits.
- Permutation: The 32-bit output is rearranged.
- Feistel operation: The output is XORed with the left half, and the halves are exchanged.

4. Final swap and permutation: After round 16, the halves are combined and the inverse initial permutation is applied to produce ciphertext.

Block diagram:

64-bit Plaintext

↓

Initial Permutation (IP)

↓

L0 (32 bits) | R0 (32 bits)

↓

16 Feistel Rounds □ 16 Round Keys

↓

Final Swap

↓

Inverse Initial Permutation

↓

64-bit Ciphertext

Decryption: DES uses the same process, but the round keys are applied in reverse order.

Limitation: Its 56-bit effective key is too short against modern brute-force attacks. Therefore, DES is considered insecure for modern applications.

Conclusion: DES is an important historical symmetric encryption algorithm, but modern systems generally use stronger algorithms such as AES.

Q2. Explain RSA algorithm with its application.

Answer:

RSA is an asymmetric-key cryptographic algorithm. It uses two different keys: a public key for encryption and a private key for decryption. Its security is based on the difficulty of factoring a large number into its prime factors.

Key generation:

1. Choose two large prime numbers p and q.
2. Calculate $n = p \times q$.
3. Calculate $\square(n) = (p - 1)(q - 1)$.
4. Choose e such that $1 < e < \square(n)$ and $\gcd(e, \square(n)) = 1$.
5. Calculate d such that $(d \times e) \bmod \square(n) = 1$.
6. Public key = (e, n); Private key = (d, n).

Encryption:

If M is the plaintext message, ciphertext is:

$C = M^e \bmod n$

Decryption:

The original message is recovered by:

$M = C^d \bmod n$

Simple numerical example:

Take $p = 3$ and $q = 11$.

$n = 3 \times 11 = 33$

$\phi(n) = 2 \times 10 = 20$

Choose $e = 3$ and $d = 7$, because $(3 \times 7) \bmod 20 = 1$.

Public key = (3, 33)

Private key = (7, 33)

For message $M = 4$:

$C = 4^3 \bmod 33 = 31$

Decrypt: $M = 31^7 \bmod 33 = 4$

Applications:

- Secure exchange of encryption keys.
- Digital signatures and verification.
- Authentication and secure communication.
- Protecting data exchanged over networks.

Note: Real RSA uses very large key sizes and secure padding schemes. The small numbers above are only for understanding.

Conclusion: RSA enables secure communication using a public/private key pair and is widely used in public-key cryptography.

Q3. Explain the concept of cryptography and illustrate its need by applying it to a real-life scenario with a suitable example.

Answer:

Cryptography is the technique of protecting information by converting readable data (plaintext) into an unreadable form (ciphertext) using an algorithm and a key. Only an authorized person with the correct key can recover the original information.

Basic terms:

- Plaintext: Original readable message.
- Encryption: Conversion of plaintext into ciphertext.
- Ciphertext: Encrypted, unreadable message.

- Decryption: Conversion of ciphertext back into plaintext.
- Key: A value used by the encryption or decryption algorithm.

Basic process:

Plaintext □ Encryption + Key □ Ciphertext

Ciphertext □ Decryption + Key □ Plaintext

Need for cryptography:

1. Confidentiality: Prevents unauthorized people from reading data.
2. Integrity: Helps detect unauthorized changes to information.
3. Authentication: Helps verify the identity of communicating parties.
4. Non-repudiation: Digital signatures can provide evidence that a message was signed by a particular key holder.
5. Secure communication: Protects information sent over public networks.

Real-life example - Online banking:

When a customer accesses online banking, sensitive information such as login details and transaction data must be protected.

- The browser and bank establish a secure connection using protocols such as TLS.
- Cryptographic techniques protect data while it travels across the internet.
- Authentication verifies the parties and helps prevent impersonation.
- Integrity checks help detect changes to transmitted data.

Conclusion: Cryptography is essential for protecting private information, securing online transactions and building trust in digital communication.

Q4. Compare the AES and DES cryptography algorithms in terms of their structure, key sizes and security features.

Answer:

AES and DES are symmetric-key encryption algorithms. Both use the same secret key for encryption and decryption, but they differ in structure, key size and security.

AES (Advanced Encryption Standard):

- Block size: 128 bits.
- Key sizes: 128, 192 and 256 bits.

- Structure: Substitution–Permutation Network (SPN).
- Rounds: 10, 12 or 14 rounds depending on key size.
- Operations include SubBytes, ShiftRows, MixColumns and AddRoundKey.
- Considered secure and widely used today.

DES (Data Encryption Standard):

- Block size: 64 bits.
- Effective key size: 56 bits.
- Structure: Feistel network.
- Rounds: 16 rounds.
- Uses expansion, XOR, S-box substitution and permutation.
- Its short key makes it vulnerable to brute-force attacks.

Comparison:

Feature	AES	DES
Block size	128 bits	64 bits
Key size	128/192/256 bits	56 effective bits
Structure	SPN	Feistel
Rounds	10/12/14	16
Security	Strong when used correctly	Insecure today
Performance	Efficient in software/hardware	Older design; less suitable today

Conclusion: AES is much more secure than DES because it supports longer keys and a larger block size. AES is preferred for modern data protection, while DES is mainly studied for historical and educational purposes.

Q5. Explain Asymmetric Key Cryptography with suitable example.

Answer:

Asymmetric Key Cryptography is a method of encryption that uses two mathematically related keys: a public key and a private key.

1. Public key: Shared openly with other users. It can be used to encrypt data or verify a digital signature.
2. Private key: Kept secret by its owner. It can be used to decrypt

data or create a digital signature.

Working:

1. The receiver generates a public and private key pair.
2. The receiver shares the public key with the sender.
3. The sender encrypts the message using the receiver's public key.
4. The encrypted message is sent to the receiver.
5. The receiver decrypts it using the corresponding private key.

Diagram:

Sender's Message
↓
Encrypt using Receiver's Public Key
↓
Ciphertext
↓ (sent over network)
Receiver decrypts using Private Key
↓
Original Message

Example:

Suppose Alice wants to send a confidential message to Bob. Bob shares his public key with Alice. Alice encrypts the message using Bob's public key. Only Bob's matching private key can decrypt it.

Applications:

- Secure key exchange.
- Digital signatures.
- Identity authentication.
- Secure email and internet communication.
- Digital certificates and TLS.

Advantages:

- No need to share the private key.
- Supports digital signatures and authentication.
- Useful for secure communication over open networks.

Limitations:

- Slower than symmetric encryption.

- Requires proper key management and trusted public-key verification.
- Incorrect implementation can create security weaknesses.

Conclusion: Asymmetric cryptography provides secure communication through a public/private key pair and is often combined with symmetric encryption for efficiency.

Q6. Explain Diffie-Hellman key exchange algorithm with its application.

Answer:

Diffie-Hellman is a key exchange method that allows two parties to establish a shared secret key over an insecure communication channel. The secret key itself is not transmitted.

Working steps:

1. Both parties publicly agree on a large prime number p and a generator g.
2. Alice selects a private number a and calculates:

$A = g^a \bmod p$

3. Bob selects a private number b and calculates:

$B = g^b \bmod p$

4. Alice sends A to Bob and Bob sends B to Alice.
5. Alice calculates the shared secret:

$K = B^a \bmod p$

6. Bob calculates the shared secret:

$K = A^b \bmod p$

Both obtain the same key because:

$K = g^{(ab)} \bmod p$

Small example:

Let $p = 23$ and $g = 5$.

Alice chooses private $a = 6$:

$A = 5^6 \bmod 23 = 8$

Bob chooses private $b = 15$:

$B = 5^{15} \bmod 23 = 19$

Alice calculates $K = 19^6 \bmod 23 = 2$.

Bob calculates $K = 8^{15} \bmod 23 = 2$.

Shared secret = 2.

Diagram:

Alice (private a) Bob (private b)

| |

Calculate A = g^a mod p Calculate B = g^b mod p

| -------- A to Bob ----------> |

| <-------- B to Alice ---------- |

Calculate K = B^a mod p Calculate K = A^b mod p

| |

Shared Secret Key is the same

Applications:

- Establishing session keys for secure communication.
- TLS and other secure network protocols.
- VPN connections and secure remote access.
- Secure messaging protocols.

Limitation: Basic Diffie-Hellman does not authenticate the parties by itself and may be vulnerable to a man-in-the-middle attack. Authentication, certificates or digital signatures are used to prevent this.

Conclusion: Diffie-Hellman enables two parties to create a common secret key without sending the key directly over the network.

Q7. Difference between Symmetric and Asymmetric Key Cryptography.

Answer:

Symmetric and asymmetric cryptography are two major methods used to protect digital information.

Symmetric Key Cryptography:

- Uses one shared secret key for encryption and decryption.
- Sender and receiver must securely share the key.
- Generally faster and suitable for large amounts of data.
- Key distribution is a major challenge.
- Examples: AES and DES.

Asymmetric Key Cryptography:

- Uses a public key and a private key.

- Public key may be shared; private key must remain secret.
- Generally slower than symmetric encryption.
- Supports digital signatures and secure key exchange.
- Examples: RSA and ECC.

Differences:

1. Keys: Symmetric uses one shared key; asymmetric uses a key pair.
2. Speed: Symmetric is faster; asymmetric is usually slower.
3. Key distribution: Symmetric requires secure sharing; asymmetric allows public-key sharing.
4. Scalability: Symmetric key management becomes difficult with many users; asymmetric simplifies key distribution.
5. Main uses: Symmetric protects bulk data; asymmetric supports key exchange, signatures and authentication.
6. Examples: AES/DES versus RSA/ECC.

Example:

A file can be encrypted using AES with a randomly generated secret key. RSA can then be used to protect that AES key for the intended receiver.

Conclusion: Both methods are important. Modern secure systems commonly combine them to achieve both speed and convenient key management.

Q8. Types of cipher.

Answer:

A cipher is an algorithm used to convert plaintext into ciphertext and recover the original message through decryption.

Main types of ciphers:

1. Substitution cipher:

In this method, each letter or group of letters is replaced by another letter or symbol.

Examples:

- Caesar cipher: Each letter is shifted by a fixed number of positions.
- Monoalphabetic cipher: One fixed substitution alphabet is used.
- Vigenère cipher: Uses multiple substitution alphabets controlled by a keyword.

2. Transposition cipher:
It changes the positions of characters without changing the characters themselves.
Examples:
- Rail Fence cipher: Characters are written in a zigzag pattern and read row by row.
- Columnar transposition: Text is written in rows and read according to a column order.

3. Block cipher:
It encrypts fixed-size blocks of data at a time.
Examples: AES and DES.

4. Stream cipher:
It encrypts data one bit or byte at a time using a generated keystream.
Example: ChaCha20.

5. Product cipher:
It combines substitution and transposition operations to provide stronger encryption. Modern block ciphers use multiple rounds of such transformations.

Conclusion: Ciphers are designed in different ways to protect information. Modern applications use strong, well-tested algorithms rather than simple classical ciphers.

Q9. Difference between block cipher and stream cipher.

Answer:
A block cipher encrypts data in fixed-size blocks, whereas a stream cipher encrypts data continuously, usually one bit or byte at a time.

Block Cipher:
- Processes fixed-size blocks of plaintext.
- Uses a key and an encryption mode to protect data.
- May require padding when the data length is not a multiple of the block size, depending on the mode.
- Commonly used for files, disks and stored data.
- Examples: AES and DES.

Stream Cipher:

- Processes data as a continuous stream of bits or bytes.
- Combines plaintext with a generated keystream, commonly using XOR.
- Usually does not require block padding.
- Useful for real-time or streaming communication.
- Example: ChaCha20.

Differences:

1. Unit of operation: Block cipher works on fixed-size blocks; stream cipher works on individual bits or bytes.
2. Padding: May be required by some block-cipher modes; generally not required for stream ciphers.
3. Speed/latency: Stream ciphers can be suitable for low-latency data streams; performance depends on the algorithm and implementation.
4. Error effect: The effect of transmission errors depends on the cipher mode and design.
5. Examples: AES/DES are block ciphers; ChaCha20 is a stream cipher.

Conclusion: Block ciphers are commonly used for bulk data and storage, while stream ciphers are useful for continuous data. The correct choice depends on the application and secure implementation.